

Introduction to Chapter 5

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Question:

What are the 3
programming
structures?



Answer:

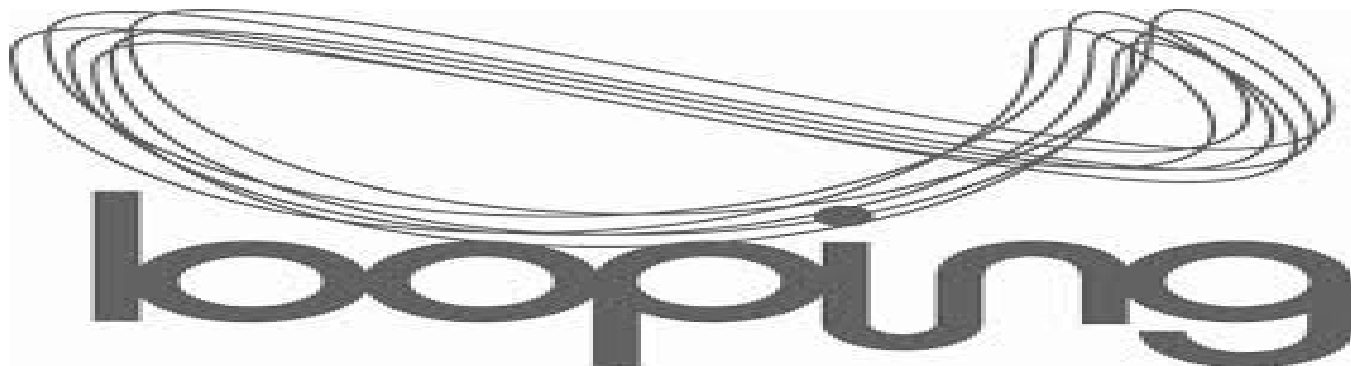
What are the 3
programming
structures?



Sequence



Selection



Selection



Decision

Example 1

```
if (condition)  
    statement;  
//endif
```

Example 2

```
if (condition)  
  { more than one statement;  
  }  
//endif
```

Question:

In the selection structure, does the *;* go at the end of the *if* statement condition?



Answer:

In the selection structure, the *;* **does not** go at the end of the *if* statement condition.



Question:

In the selection structure, does the *;* go at the end of the *each* statement after the condition?

```
if ( the shirt is red)
    wash it;  
//endif
```

```
if (count < 10)
    cout<<"The count is " << count;  
  
//endif
```

Answer:

In the selection structure, the *;* goes at the end of the *each* statement after the condition.



This is the
condition
statement.

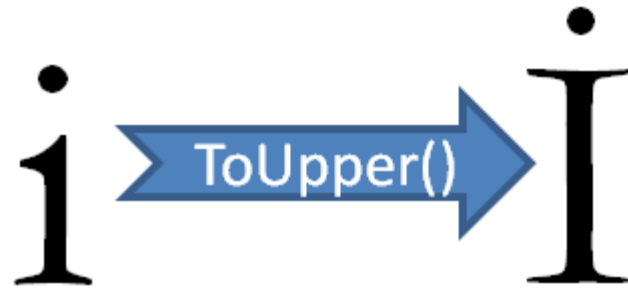
```
if (count < 10)
{   count = count +1;
    cout<<"The count is " << count;
}
//endif
```

toupper or tolower

- Converts a character variable (char) to upper or lower case, no matter what the user enters on the screen.
- Functions
- Example
 - toupper(letter)
 - tolower(letter)
- Example
 - toupper(variable name)
 - tolower(variable name)

Question:

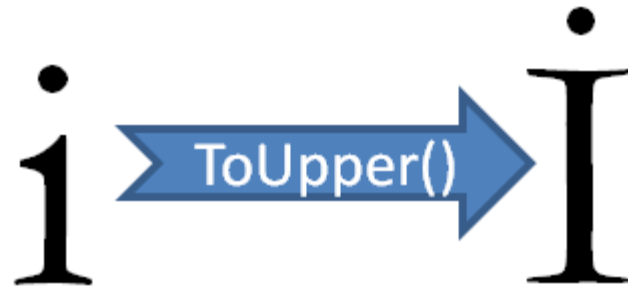
Why use
toupper or
tolower?



The diagram illustrates the transformation of a lowercase character to its uppercase equivalent. On the left is a lowercase 'i'. A blue arrow points from this 'i' to an uppercase 'I' on the right. The text 'ToUpper()' is written inside the blue arrow, indicating the method used for the conversion.

Answer:

To have
control over
incoming
data.



If the user enters a letter on the screen, do you want to have to ask every time if it is a capital or lowercase?

Use toupper or tolower to make comparisons easier.

```
#include  
<iomanip>
```

- This is a new directive to put in your program when you are going to use `setprecision` or fixed or scientific stream manipulators.

setprecision & fixed

- Are stream manipulators
- Allows you to control the number of decimal points displayed on the screen.

```
double sales = 3500.6;
```

```
cout <<fixed;
```

```
cout <<setprecision (2);
```

```
cout<< sales;
```

Sales will display as 3500.60

Question:

If you have more than one statement under the if condition, what else is required?

```
if (sales <= 16.67)
```

```
    sales = 16.67;
```

```
    Cout << "You did not make many  
    sales!" << endl;
```


Answer:

You need
brackets!

```
if (sales <= 16.67)
{
    sales = 16.67;
    Cout << "You did not make many
sales!" << endl;
}
```

Question:

Which of the following statements is correct?

Example A

```
if (a = 100)
    cout <<
    "Great Job!";
//endif
```

Example B

```
if (a == 100)
    cout <<
    "Great Job!";
//endif
```

Answer:

Which of the following statements is correct?
B, because it is a comparison.

Example A

```
if (a = 100)
    cout <<
    "Great Job!";
//endif
```

Example B

```
if (a == 100)
    cout <<
    "Great Job!";
//endif
```